



Computing CMB temperature fluctuations for spherical spaces

Preparation Bachelor's Project

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Abstract

The study of temperature fluctuations in the cosmic microwave background (CMB) gives a way to understand the possible topology of the universe. In this paper, we study how assuming that the universe has a finite **spherical topology**, can constrain the possible allowed temperature fluctuations of the CMB, resulting in distinct theoretical predictions for the temperature **anisotropies**. We begin by introducing a foundation for analyzing spherical 3-manifolds, focusing on **spectral geometry**, **eigenmode analysis**, and the **Laplace-Beltrami operator** to describe these temperature fluctuations. Then, following the results from the existing literature, we take a look at how different specific quotient spaces of the 3-sphere—such as **lens spaces** and **isospectral but non-isometric manifolds**—restrict the allowed eigenmodes of the CMB fluctuations. Next, we study the implications of inhomogeneous spherical spaces, where CMB fluctuations depend on observer location. Finally, to verify the viability of alternative cosmological models, we analyze **statistical tests for CMB** temperature fluctuations in harmonic space using real data from missions such as the WMAP.

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Cover letter

Suggestions we implemented:

1. We changed the order in which we introduce the concepts such as the quotients of the sphere and the Laplace-Beltrami operator. They are now introduced at the beginning of the article, instead of multiple times throughout the text.
2. We modified the tone of the abstract to be more formal and put it on a separate page.
3. We changed section 4.3 to be more cohesive and have an actual conclusion, instead of ending abruptly.
4. We fixed a multitude of typos and grammatical errors.
5. We removed unnecessary usage of parentheses.
6. We removed a paragraph that appeared twice.
7. We added an explanation of what eigenmodes are.

Suggestions we did not (completely) implement:

1. Some referee reports have pointed out that the lack of an explanation / motivation for specific physics formulae, formulae that weren't covered in earlier courses that are part of our bachelor program. We agree that such formulae should be explained, but detailed explanations and derivations of every formula used would force us to exceed the word limit substantially, obscuring the main ideas, thus being unfeasible.

1 Introduction

The *cosmic microwave background* (CMB) is the electromagnetic radiation left from the Big Bang. Temperature fluctuations provide an idea of the early state of the universe, as well as possible hints on its current shape. Observations of space missions such as COBE, WMAP and PLANCK have revealed an unexpectedly low variance in *CMB anisotropies*, which are small variations in the radiation, at very large angular scales. Angular scale refers to the measure of separation between points on the celestial sphere, typically expressed in degrees or radians. Low variance in anisotropies goes against the expected results of the infinite and flat universe from the current standard model of the universe, i.e. the *Lambda Cold Dark Matter model* (Λ CDM model).

This motivates exploring cosmological models beyond infinite flat spaces. One possible explanation is that the universe is not infinite, as often assumed, but finite and multiconnected, that is connected but not simply connected. Specifically, considering the universe to be a finite spherical space (positively curved and closed, rather than infinite and flat) would naturally introduce an upper limit to the size of CMB fluctuations [2][3], potentially explaining lower correlations at large angular scales.

The spherical space topology has been studied as a candidate for the shape of the universe by many researchers. For example, the 3-sphere $\mathbb{S}^3$ and its quotient manifolds can produce lower CMB power on large scales due to their finite size [3]. Various studies have systematically examined families of spherical topologies, such as lens spaces $\mathbb{S}^3/Z_p$, where the quotient of $\mathbb{S}^3$ is taken by the group of rotations of order p , with $p \leq 500$ being a typical upper bound [2]. While such models did not show a dramatically strong suppression of CMB power in the tested range, they set an important methodology for how one might analyze and compare CMB temperature fluctuations in spherical space models. The important component of the articles involved is how these theoretical results compare to observations and what are the implications of these findings regarding the possible topology (i.e. shape) of the universe.

2 Preliminaries

2.1 Manifolds

Metric and topological spaces allow us to generalize concepts from analysis, including limits, sequences, and continuity. As a reminder, metric spaces generalize distance and topological spaces generalize neighborhoods, which can both be used to reconstruct the concepts from analysis. This is useful, as it allows those concepts to be used in novel settings.

The definition of a topological space is too general to meaningfully describe physical reality. The concept of topological manifolds tries to limit the potential pathological examples of topological spaces by requiring the spaces to look Euclidean at the local level. Formally, this means that any point on the set has a neighborhood that is equivalent (homeomorphic) to an open set of Euclidean space.

Definition 2.1 (Topological manifold). A topological space M is a topological manifold of dimension n , or topological n -manifold, if it has the following properties:

- M is a Hausdorff space,
- M is second countable,
- M is locally euclidean of dimension n , that is, for any point $p \in M$ there exists an open subset $U \subset M$, a point $p \in U$, and an open subset $V \subset \mathbb{R}^n$ together with a homeomorphism $\varphi : U \rightarrow V$.

We recall that second-countability is defined as follows:

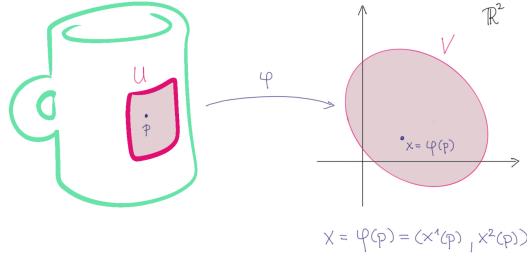


Figure 1: The prototypical example of a manifold – a mug. Image source: [11]

Definition 2.2 (second countable). A topological space $(X, \mathcal{T})$ is said to be second countable if there exists a countable set $B \subset \mathcal{T}$ such that any open set can be written as a union of sets from B . In such case, B is called a (countable) basis for the topology $\mathcal{T}$.

For one, limits need not be unique in the general case. This could happen if there are not enough neighborhoods to direct the sequence to a single point, which is why we require manifolds to be Hausdorff.

Moreover, sequences are just functions from the naturals to our space, so they are inherently *countable* in nature. Indeed, we are used to many nice properties of limits. For instance, if a function in real space preserves the limits of sequences, then it is guaranteed to be continuous. But such a test might not hold in the general case. Indeed, we could find too many neighborhoods around each point, such that a simple sequence does not suffice for detecting properties like continuity. Requiring the space to be *first countable* (that is, requiring that each point has a countable basis for its neighborhoods) solves this issue. In fact, we require an even stronger condition for manifolds in second countability (that is, a countable basis must exist for the entire space).

For a more in-depth treatment of manifolds and their many-fold properties, the reader should consider [11]. A more thorough exposition of the way limits and sequences break down in the general setting (and how we can work around that) can be found in the third chapter of [12].

2.2 Curvature and holes

When transporting (i.e. moving) a vector around a loop on a flat plane, the vector will arrive at the starting point in the same orientation it left as. This turns out not to be the case in the more general setting. We can quantify the amount by which this fails around every point using the concept of Gaussian curvature. Intuitively, when the curvature is 0, the space is flat (at a point). When the curvature is positive, the space is convex (around said point), giving rise to spherical geometry. Otherwise, the space is hyperbolic (around the point). For larger dimensions, the Riemann curvature tensor allows us to take the Gaussian curvature along different slices formed by different sections.

A natural application of the above is detecting the curvature of the universe itself, i.e. the nature of the (spatial) geometry in which we exist. Our best tool when it comes to trying to detect such matters is the study of the cosmic microwave background radiation. In particular, this article is interested in the consequences of assuming that the underlying geometry is spherical. Indeed, although this might sound trivial (i.e. the universe would spatially be the surface of a giant four-dimensional ball), more interesting spaces arise when we allow the underlying space to have *holes*.

Topologically, holes can be detected using homotopy groups (the group of loops in the space, under the identification of loops that can be smoothly transformed into each other as equivalent).

In particular, a space is contractible if any loop can be smoothly transformed into the trivial loop (a path that never leaves the initial point). That is, the (first) homology group must be trivial. We say a space is simply connected when it is both contractible and path connected (i.e. every two points are connected by a path).

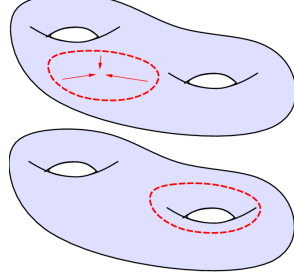


Figure 2: Diagram showing two double tori. The red dotted path on the top double torus is a contractible loop, while the red path on the bottom one is not contractible. [?]

2.3 Quotients of the 3-sphere

In what follows, we use the concrete definition of the 3-sphere as an embedded submanifold of $\mathbb{R}^4$ lying in Euclidean space, namely $\mathbb{S}^3 := \{x \in \mathbb{R}^4 \mid \|x\| = 1\}$. Next we define a special set of matrices which induces well-defined maps on the sphere.

Definition 2.3. We define

$$SO(4) := \{A \in \mathbb{R}^{4 \times 4} : A^T A = I_4, \det(A) = 1\}.$$

By the standard action of $\mathbb{R}^{4 \times 4}$ on $\mathbb{R}^4$, we identify $SO(4)$ naturally as a subgroup of the isometry group of $\mathbb{S}^3$. This is because for any $x \in \mathbb{S}^3$ we get

$$\|Ax\| = (Ax)^T(Ax) = x^T(A^T A)x = x^T x = \|x\| = 1$$

hence $Ax \in \mathbb{S}^3$ implying $A|_{\mathbb{S}^3}$ is a well-defined map and a bijection. By analogous reasoning, we find that

$$\|Ax - Ay\| = \|A(x - y)\| = \|x - y\|,$$

hence A is an isometry. Next, for any subgroup $\Gamma \leq SO(4)$ we can consider the orbits of the action of Γ on $\mathbb{S}^3$. We recall from group theory that this produces an equivalence relation on $\mathbb{S}^3$, where $x \sim y$ precisely when x and y are in the same orbit of Γ . That is, the set of equivalency classes of $\mathbb{S}^3$ with

$$x \sim y \iff \exists g \in \Gamma : x = g \cdot y.$$

The set $\mathbb{S}^3/\sim$ can be given a manifold structure if certain conditions on the projection $\pi : \mathbb{S}^3 \rightarrow \mathbb{S}^3/\sim$ are satisfied, the details of which can be found in [11]. For instance, if Γ is a finite subgroup of $SO(4)$ then the manifold $\mathbb{S}^3/\sim$ is a 3-manifold with positive curvature, similarly to $\mathbb{S}^3$ itself (which can be constructed when starting with the trivial group).

3 Discussion of results from selected articles

3.1 The spectral geometry of hyperbolic and spherical spaces

In [9], the authors go over the theoretical grounding of the literature on the topic. We refer to quotients $\mathbb{S}^n/\Gamma$, defined analogously to the $\mathbb{S}^3$ case for some subgroup Γ of the isometry group of the n -sphere as *sphere forms*.

An important class of manifolds are the so-called *lens spaces*, which correspond to cyclic groups Γ with the dimension of the underlying manifold being odd. As we are interested in studying the shape of our (spatially) three-dimensional universe, all the spaces we are interested in will be lens spaces as long as Γ is cyclic.

Definition 3.1 (Lens spaces). Let $q \in \mathbb{N}$ and $s \in \mathbb{Z}^n$ such that $\gcd(q, s_i) = 1$ for all i . We then defined the lens space $L(q, s) := \mathbb{S}^{2n-1}/\Gamma_{q,s}$, as the space induced by $\Gamma_{q,s}$, the (cyclic) group generated by the diagonal block matrix with entries¹ $R(2\pi s_i/q)$.

The condition $\gcd(q, s_i) = 1$ is necessary to ensure that Γ acts freely on $\mathbb{S}^{2n-1}$, i.e. for every $g \in \Gamma$ and $x \in \mathbb{S}^{2n-1}$ the relation $gx = x$ implies $g = \text{id}$, or in other words, the action of g has no fixed points for any $g \neq \text{id}$.

The *Laplace-Beltrami operator* generalizes the Laplace operator from smooth functions defined on $\mathbb{R}^n$ to general manifolds.

Definition 3.2 (Laplace-Beltrami operator). The Laplace-Beltrami operator is defined as the divergence of the gradient

$$\Delta f := \text{div}(\nabla f).$$

The *spectrum* of the Laplace-Beltrami operator is defined as its collection of eigenvalues. We also refer to this spectrum as the spectrum of the underlying manifold. It turns out that said spectrum is a discrete subset of the non-negative reals, in which every value occurs with a finite multiplicity. Two manifolds are said to be *isospectral* if they share a spectrum.

The *Inverse spectral geometry* studies the different ways in which the spectrum affects different properties of the underlying space. For instance, the dimension and volume of a manifold are completely determined by its spectrum. However, the spectrum does not completely determine the manifold uniquely up to isometry.

3.1.1 Volume maximizing isospectral pairs of sphere forms

The first important problem tackled by [9] is that of determining the *isospectral sphere form pair* (that is, a pair of spaces that are isospectral yet not isometric) of largest volume. According to [6], if $\mathbb{S}^{2n-1}/\Gamma_1$ and $\mathbb{S}^{2n-1}/\Gamma_2$ are isospectral spherical forms, the two groups have equal order and equal set of element orders. As a consequence, one is cyclic only precisely when the other is cyclic (thus one space is a lens space if and only if the other is as well).

After classifying Γ for spherical forms with non-cyclic homotopy group of order strictly smaller than 24 into four specific groups of different orders (the specific groups are irrelevant here), the paper concludes that if a pair of odd-dimensional isospectral but not isometric spherical space forms exists with fundamental group of order strictly lesser than 24, then the two are lens spaces, otherwise one of the two would be one of the four non-cyclic groups classified above, but then so would the other, as by [6] isospectral pairs share their cyclic-ness. The isospectral pair with the largest volume must then also be the one containing lens spaces, since no other kind of pair can exist for such groups. The paper then defines a way to generate larger lens spaces from smaller ones by concatenating the indices, which we denoted earlier by s , and can be treated like ordered lists here. Together with a computational search, the paper was able to build a table containing the precise lens spaces that maximize volume for multiple dimensions, although whether lens spaces are the solutions for all dimensions is still an open question.

¹Here we use $R(\theta)$ to denote the 2-dimensional rotation matrix rotating by the given angle, i.e.

$$R(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

3.1.2 The silver lining

While the study of isospectral pairs is interesting in its own right, we can take solace in knowing that no such pairs exist in dimension three. That is, for the purpose of this article, all pairs of isospectral sphere forms must also be isometric. The motivated reader is encouraged to read [9] for further details. For the rest of this paper, our primary means of CMB analysis will relate to the spectrum of different concrete models of spherical universes.

A key aspect of this pursuit is understanding the ways in which *eigenmodes of the Helmholtz equation* (which we formally define in the next section) behave under different spherical topologies, as these modes determine the possible temperature fluctuations in a CMB model. The work of Aurich, Lustig, and Steiner (2005) explores these spectral properties and their consequences for modeling CMB fluctuations in spherical space forms.

3.2 CMB Anisotropy of Spherical Spaces

In [3] the authors consider the 3-manifolds of the form $M := \mathbb{S}^3/\sim$ where $\sim$ is an equivalence relation coming from identifying orbits of some finite subgroup H of the special orthogonal group $SO(4)$.

On any smooth manifold one can define the notion of smooth functions and the Laplace-Beltrami operator Δ , as explained in the previous section. One can moreover consider the solutions of the so-called *Helmholtz equation* on M given by

$$(\Delta + E_\beta^M)\psi_\beta^{M,i} = 0,$$

where $E_\beta^M \in \mathbb{R}$, the function ψ_β^M is a smooth function on M , and $\beta \in \mathbb{N}$. We call the function ψ_β^M the *eigenmode* of the Helmholtz operator and E_β^M the corresponding eigenvalue. As explained in [10], we can lift any eigenmode ψ_β^M of the Helmholtz equation on M to an H -invariant eigenmode $\psi_\beta'^M$ on $\mathbb{S}^3$, meaning it satisfies the Helmholtz equation and $\psi_\beta'^M(hx) = \psi_\beta'^M(x)$ for all $h \in H$.

From [3] we know that the spectrum of the Helmholtz operator is a discrete subset of $\mathbb{R}$ and that $E_\beta = \beta^2 - 1$ for $\beta \in \mathbb{N}$. We call β a *wave number*. However, depending on the group H , not every natural β is a possible wave number — the eigenmode does not exist for some. For example:

1. If $H = \{1\}$ (the trivial group), the set of wave numbers is $\mathbb{N}$ itself without restrictions.
2. If $H = \mathbb{Z}/m\mathbb{Z}$ for odd m (that is, if the underlying manifold is a lens space), the set of wave numbers is

$$\{1, 3, \dots, m\} \cup \{m+1, m+2, m+3, \dots\}.$$

3. If $H = O^*$, the binary tetrahedral group — a subgroup of $SO(4)$ of size 24, the set of wave numbers turns out to be

$$\{1, 7, 9\} \cup \{13, 15, 17, \dots\}.$$

Furthermore, we can write any eigenfunction of the Helmholtz operator on M in terms of known eigenfunctions on $\mathbb{S}^3$ while being able to compute the necessary coefficients.

Section 3 of [3] constitutes the technical part of the paper, the complete understanding of which requires a background in cosmology. The authors' goal is to compute the relative fluctuations $\frac{\delta T}{T}$ of the CMB for the various spherical manifolds obtained as the quotients of finite subgroups of $SO(4)$. First, they explain how the CMB fluctuations arise as manifestations of different effects, such as the Sachs-Wolfe contribution. For the computations, the authors assume that the initial values for the fluctuations corresponding to each wave number are certain Gaussian random variables.

The authors arrive, for each spherical manifold, at the value for the *angular power spectrum* δT_l^2 when $l = 1, 2, 3$, which are real numbers that can be directly derived from δT written as

$$\delta T(\hat{n}) = \sum_{l \geq 2} \sum_{m=-l}^{m=l} a_{lm} \tilde{Y}_{lm}(n)$$

in terms of $\tilde{Y}_{lm}$ which are *spherical harmonics* - a certain family of smooth functions defined on $\mathbb{S}^2$ which forms an orthonormal basis for $L^2(\mathbb{S}^2)$, meaning they are orthogonal and of size 1, with respect to the L^2 inner product. The two sphere $\mathbb{S}^2$ is the natural domain for the CMB when we observe it from Earth.

The angular power spectrum is defined as

$$\delta T_l^2 = \frac{l(l+1)}{2} \frac{1}{2l+1} \left\langle \sum_{m=-l}^l a_{lm}^2 \right\rangle.$$

For each spherical manifold, the authors plot δT_l^2 with respect to Ω_{tot} , together with δT_l^2 as measured in WMAP. The variable Ω_{tot} is the density parameter of the universe, and can be obtained from measurements.

3.2.1 Results

The authors conclude that the spherical manifolds which align closest with the measured data are the ones obtained when setting $H = O^*$ and $H = I^*$, the *binary octahedral* and *binary icosahedral* groups respectively, which are of orders 48 and 120. The authors present an explicit description of the groups generated by left- and right- multiplication of certain quaternions, but we will omit the details for the sake of brevity.

Furthermore, the authors excluded other groups (such as the binary tetrahedral group T^* and infinite families of binary dihedral groups and cyclic groups) as possible models of the universe, as they exhibit drastically different behavior.

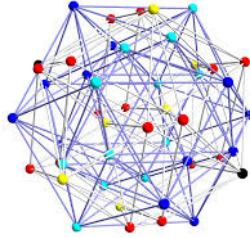


Figure 3: Graphical representation of the binary tetrahedral group [?]

To conclude, the above illustrates how different spherical spaces influence CMB temperature fluctuations through their spectral properties and eigenmodes. However, some of the models discussed thus far assume a level of uniformity in the way spaces behave. In reality, some spherical manifolds are *inhomogeneous* — their CMB temperature distributions can depend on the observer's location. To account for this, we now examine [1], which analyses the effects of inhomogeneity on CMB radiation, focusing on how different observer positions within these spaces influence measured fluctuations.

3.3 CMB radiation in an inhomogeneous spherical space

In [1], the authors consider the CMB in spherical 3-manifolds that are *multi-connected* and inhomogeneous, comparing them with their homogeneous counterparts. To start off, the manifolds used in the paper are all of the form $\mathbb{S}^3/\Gamma$, where Γ is a group acting on $\mathbb{S}^3$, with the quotient being the set of orbits. The space being multi-connected means it is not simply connected, i.e. that not all loops are contractable to a point.

The condition of the manifold being inhomogeneous means that, unlike the 3-sphere $\mathbb{S}^3$ and 3-torus $\mathbb{T}^3$, the manifold does not look the same at each point on the manifold. We will define this properly later. Because of inhomogeneity, the CMB is very much dependent on the point of observation.

An important concept needed for formalizing this concept is that of the fundamental domain. Given the manifold $\mathbb{S}^3/\Gamma$, a fundamental domain is a subset of $\mathbb{S}^3$ which covers the entirety of $\mathbb{S}^3$ under the action of Γ . Moreover, all non-identity elements of Γ must send elements in this set outside of it.

Intuitively, one can imagine the manifold being covered by disjoint copies of the fundamental domain placed next to each other, similarly to how $\mathbb{R}$ is tiled by the intervals $[n, n+1)$ for $n \in \mathbb{Z}$, forming a collection of such domains that cover the space under the standard action of $\mathbb{Z}$ on $\mathbb{R}$. Importantly, we are interested in one specific fundamental domain called the *Voronoi domain*. Given an observation point $x_0 \in \mathbb{S}^3$, the Voronoi domain is the set of elements $x \in \mathbb{S}^3$ such that

$$d(x_0, x) \leq d(x_0, g \cdot x) \quad \forall g \in \Gamma,$$

where $d(\cdot, \cdot)$ is the $\mathbb{S}^3$ -distance, analogous to the angular distance on $\mathbb{S}^2$. We define a manifold to be *homogeneous* if the Voronoi domain is identical regardless of the observer $x_0 \in \mathbb{S}^3$. Otherwise, the manifold is called *inhomogeneous*.

To determine whether a spherical manifold is homogeneous, one must understand the way group action changes with observer position. Changing the observer position is done mathematically by applying a change of coordinates $u' = uq$ with $u \in \mathbb{S}^3$ and q an isometry in $\text{SU}(2, \mathbb{C}) \equiv \mathbb{S}^3$, making $u = q^{-1}$ the new origin $u' = q^{-1}q = e$. Then, using the general group element $g = (g_l, g_r) \in \Gamma$ and letting $\tilde{u} = g \cdot u = (g_l)^{-1}ug_r$, we get $\tilde{u}' = (g_l)^{-1}u'(q^{-1}g_rq)$. The transformation in our shifted space is then $g' = (g_l, q^{-1}g_rq)$, which generally differs from $g = (g_l, g_r)$. Therefore, if $g_r = q^{-1}g_rq$ does not hold for some g and q in the manifold, then the manifold is known to be inhomogeneous.

In the article, the authors discuss different spherical 3-manifolds, classifying whether they are inhomogeneous, analyzing their properties, and finding equivalences. Specifically, they look at manifolds of the form $\mathbb{S}^3/\Gamma$ with Γ a group of order 8, in order to make the volume equal to $2\pi^2/8$. Within those restrictions, we have the lens spaces $L(8, 1)$ and $L(8, 3)$, the cubic Platonic manifolds $N2$ and $N3$, and the manifold D_8^* obtained when Γ is the binary dihedral group. Analyzing said manifolds, one finds the equivalences $N2 \equiv L(8, 3)$ and $N3 \equiv D_8^*$. Furthermore, using the theory introduced previously, we learn that $N2$ is inhomogeneous while $N3$ and $L(8, 1)$ are homogeneous.

3.3.1 Results

The rest of the article is about finding out how the CMB would look like in our three spaces $L(8, 1)$, $L(8, 3) \equiv N2$ and $N3$, with a focus on the difference between the homogeneous and inhomogeneous manifolds. For example, it finds that inhomogeneous spaces like $N2$ have a much higher variety of CMB anisotropies, because of how elements $g \in \Gamma$ act different depending on the observer. The main result is related to the suppression of CMB anisotropy over angles 60° and larger. We see the strongest suppression in the Voronoi domains of $N2$ that are Platonic cube shaped. On the other hand, we see the least suppression in the lens shaped Voronoi domains of $N2$. Looking at $N3$ together with the total energy density parameter $\Omega_{\text{tot}} > 1.07$, we notice an even larger suppression than for observers in $N2$.

Given the potential for anisotropies in spherical space models, it is crucial to test whether CMB fluctuations exhibit directional dependence. The study by [7] develops a statistical framework for detecting anisotropies in the mean CMB temperature fluctuations. Their method employs spherical harmonic decomposition and Monte Carlo simulations to compare theoretical predictions with WMAP observational data.

3.4 Test for anisotropy in the mean of the CMB temperature fluctuation in spherical harmonic space

The temperature fluctuations of the Cosmic Microwave Background (CMB) provide crucial insights into the physics of the early universe and the underlying cosmological model. A fundamental assumption in standard cosmology (Λ CDM model) is *statistical isotropy*, meaning that the statistical properties of the CMB should be the same in all directions. Testing this assumption is essential, as deviations from isotropy could indicate alternative topologies for the universe.

Following the article by [7], this section goes through an overview of a mathematical method for computing and interpreting CMB temperature fluctuations in spherical spaces, incorporating statistical isotropy, covariance structures, and Monte Carlo simulations, which are later compared with the Wilkinson Microwave Anisotropy Probe (WMAP) seven-year observation data.

From the perspective of an Earth-based observer, we can view CMB as a function defined on the celestial sphere $\mathbb{S}^2$. Hence we can express CMB as the sum of spherical harmonics. The observed temperature fluctuations along a given direction in the celestial sphere can be expanded using spherical harmonics

$$\Delta T(\hat{n}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\hat{n}),$$

where $\Delta T(\hat{n})$ are fluctuations on the CMB temperature field in the direction of $\hat{n}$, $Y_{\ell m}$ are the spherical harmonic functions - which form an orthonormal basis for smooth functions on the sphere, and $a_{\ell m}$ are the *spherical harmonic coefficients* (encoding temperature fluctuations at different angular scales).

Under the assumption that the primordial fluctuations are statistically homogeneous and isotropic in the mean, and assuming they obey the Gaussian distribution, with the $2\ell + 1$ spherical harmonic coefficients $a_{\ell m}$ for each ℓ being independent Gaussian variables, then the coefficients should satisfy the following condition:

$$\langle a_{\ell m} \rangle = 0, \quad \langle a_{\ell m} a_{\ell' m'}^* \rangle = C_{\ell} \delta_{\ell \ell'} \delta_{m m'}.$$

Here $\langle \dots \rangle$ denotes the average, C_{ℓ} is the ensemble average *power spectrum*, which describes the variance of temperature fluctuations as a function of angular scale, and δ is the Kronecker delta. The condition $\langle a_{\ell m} \rangle = 0$ implies that the mean of the CMB temperature fluctuations should be zero when averaged over many realizations. Thus, if the mean is nonzero $\langle a_{\ell m} \rangle \neq 0$, it **suggests possible anisotropies or preferred directions in the universe** [7].

Since real CMB observations are affected by instrumental noise and *sky masking* (regions obscured by the Milky Way), estimating C_{ℓ} accurately requires simulations. To test for anisotropy, researchers compare the observed mean harmonic coefficients with theoretical predictions using Monte Carlo methods. Given a theoretical power spectrum C_{ℓ}^{th} , used to generate synthetic realizations of the CMB sky, the corresponding harmonic coefficients are drawn from a Gaussian distribution:

$$a_{\ell m} \sim \mathcal{N}(0, C_{\ell}^{\text{th}}).$$

Applying a *sky mask function* $M(\hat{n})$ modifies the observed coefficients:

$$a_{\ell m}^{\text{mask}} = \sum_{\ell' m'} M_{\ell m, \ell' m'} a_{\ell' m'}^{\text{all sky}},$$

where $a^{\text{all sky}}$ represents the full-sky harmonic coefficients (which are unobtainable in practice), and $M_{\ell m, \ell' m'}$ is the convolution matrix of the mask, which accounts for mode coupling due to masking effects. However, this process introduces bias, which must be corrected using a decorrelation transformation based on the eigenvalue decomposition of the covariance matrix.

Before we proceed, it is important that we introduce the concept of a *multipole*, this corresponds to a mode in the spherical harmonic expansion of the CMB temperature fluctuations. Where each multipole represents a different angular scale in the sky. Moreover, we also have the *multipole moment* ℓ , which corresponds to angular scales.

To test the assumption of statistical isotropy, we examine the mean of $a_{\ell m}$ across multipole ranges. The test statistic is defined as:

$$S_i = \sum_j W_{ij} M_j,$$

where W is a decorrelation matrix derived from the covariance structure. If significant anomalies appear in the mean values of $a_{\ell m}$, this suggests deviations from isotropy that may indicate the need for new theories.

For plotting, we used the results of different frequency channels used in the CMB analysis, originating from the WMAP (Wilkinson Microwave Anisotropy Probe) data. This corresponds to the following frequency band categories: Q-band (40GHz), V-band (60GHz) and W-band (90GHz) [5]. Each band has slightly different noise levels and sensitivity to galactic foregrounds.

Taking these data, we plot the comparison between expected noise levels and observed anomalies in the mean multipole moments. Let the x -axis represent the multipole moment ℓ , which corresponds to angular scales, with lower values representing large-scale features and higher values representing smaller-scale fluctuations. The y -axis represents the test statistic S_i , which quantifies deviations from statistical isotropy, or in other words the temperature fluctuations from the expected results for an isotropical space.

The multipole moment ℓ and the angular scale θ are inversely related. The higher the value of ℓ , the smaller the angular scale, meaning we are looking at finer details. This relationship is approximately given by: $\theta \approx \frac{180^\circ}{\ell}$.

Plotting the results for the frequency bands Q, V and W , we get the graph for the decorrelated Statistical Spectrum in Figure 3 (This is the main graph from [7])

We should note that Figure 3 contains a bin of size 20. The bin size in this context refers to the range of multipole moments ℓ that are grouped together when calculating the test statistic S_i . Or, in other words, instead of analyzing each individual ℓ separately, we sum over a range of values (a bin). Moreover, Figure 3 presents the plots for:

- $Q + V + W$: A combined dataset that merges all three frequency bands to improve the signal-to-noise ratio.
- $Q1 - Q2, V1 - V2, W1 - W2$: These are difference maps, created by subtracting different detector readings of the same frequency band. Due to this, they primarily contain instrumental noise, since the real CMB signal cancels out when differencing measurements from the same frequency.

These maps help verify whether anomalies seen in the main dataset are truly cosmological or just instrumental artifacts.

Now, to understand this graph, note that: $S_i > 0$ represents more temperature fluctuations than expected at that scale. Therefore, $S_i < 0$ corresponds to fewer temperature fluctuations than expected on that scale. In short, larger absolute values of S_i correspond to a stronger

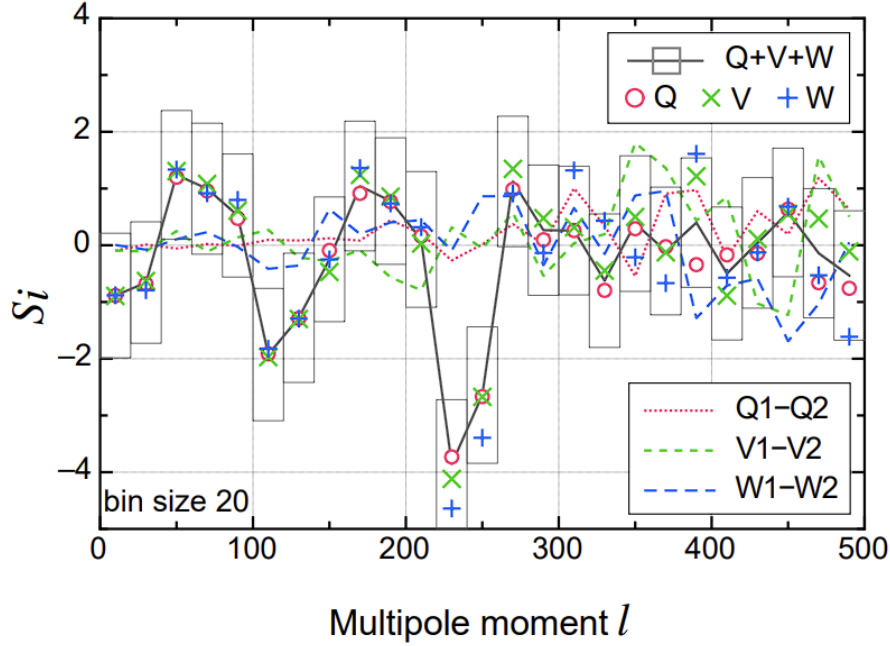


Figure 4: The decorrelated band mean spectrum obtained by the seven-year WMAP data. The boxes and solid line are the result of the overall combined map ($Q + V + W$) and color data points are the results of the individual frequency maps (Q, V, W). The horizontal width of boxes indicates the bin size $\Delta\ell = 20$ for calculating the binned mean spectrum. Image source: [7]

deviation from isotropy. Moreover, to understand the multipole moment ℓ and what it represents in the graph, we can break it down into three main scenarios:

- Deviations only at high ℓ (small scales, $\ell > 300$): This would correspond to deviations caused by instrumental noise or foreground contamination [7].
- Deviations only at low ℓ (large scales, $\ell < 50$): This would correspond to deviations caused by large-scale anisotropies, potentially challenging the assumption of a statistically isotropic universe [4][8].
- Deviations on both large and small scales: This could indicate a more fundamental cosmological effect affecting multiple scales.

In Figure 3 we can see that the most significant anomaly appears in the mid-range multipoles ($221 \leq \ell \leq 240$), where S_ℓ is strongly negative, suggesting that the mean spherical harmonic coefficients at this scale are lower than expected compared to a statistically isotropic model. In other words, this means that temperature fluctuations at these angular scales are weaker than predicted by standard cosmological models. Thus, we can say that these results suggest something unexpected at these angular scales. But what is the cause? we can take a look at the three main scenarios stated above for the multipole moment ℓ ranges. In our graph, we account for instrumental noise with the plots for $Q1-Q2$, $V1-V2$, $W1-W2$. Since the anomalies appear in $Q + V + W$ but not in $Q1-Q2$, $V1-V2$, or $W1-W2$, it suggests they are not due to noise, but instead a real cosmic effect. Moreover, the significant anomaly occurs at $221 \leq \ell \leq 240$, not at high ℓ ($\ell > 300$), where the noise-dominated behavior appears. This means that instrumental noise alone cannot explain the observed anomaly. Similarly, the strongest deviation we observe

is at intermediate scales $221 \leq \ell \leq 240$, not at very large scales ($\ell < 50$). Thus, our anomaly does not directly suggest a large-scale anisotropy. If we try to look at deviations at both large and small scales simultaneously. In this case, the main anomaly is localized in a particular range of ℓ , rather than appearing at both low and high ℓ . So, this suggests that it is not a fundamental cosmological effect affecting multiple scales.

The presence of an anomaly at $221 \leq \ell \leq 240$ suggests a possible connection to models of a finite curved universe. In a positively curved space (e.g., a three-sphere), allowed wave modes are quantized, which could lead to certain multipoles scales being preferred or suppressed in the CMB power spectrum. The results of this article, although not conclusive, support the idea that the universe (or at least the CMB sky) has a small preferred direction. Supporting the idea of a non-trivial topology in the universe, specifically a spherical one. Which is analyzed more into detail in the other articles discussed in the other sections. As a final note, the analysis is limited by the existing data for the CMB temperatures. Thus, future high-resolution independent CMB observations like PLANCK will further refine these methods, offering deeper insights into the fundamental nature of the universe and the topology of space-time.

4 Conclusion

The study of CMB temperature fluctuations provides a means of establishing the geometry and topology of the universe. By analyzing key results from various studies, we discovered how different finite spherical topologies can constrain the possible allowed temperature fluctuations of the CMB, resulting in distinct theoretical predictions for the temperature anisotropies.

We took a detour into the realm of *inverse spectral geometry and the Laplace-Beltrami operator*, and investigated isospectral pairs of spherical forms. Were any such pairs to exist, ambiguity in CMB analysis would become an issue, as our primary approach is analysis of the manifolds' spectrums. The silver lining is that all three-dimensional isospectral pairs of spherical spaces are also isometric, implying that determining the eigenmode spectrum is enough to determine the underlying topology for our spatial universe.

We then examined how different quotient spaces of the 3-sphere ($\mathbb{S}^3/\Gamma$), align with the CMB anisotropies. We found that certain spherical manifolds can lead to **suppressed large-scale CMB fluctuations**, potentially addressing anomalies observed in WMAP and Planck data. Moreover, the **binary octahedral and binary icosahedral** spaces have power spectra that align more closely with observed CMB anomalies, whereas other topologies (e.g., binary tetrahedral and cyclic groups) were found to be inconsistent with observations. We then examined the impact of **inhomogeneous spherical spaces**, where CMB temperature fluctuations may depend on the observer's location. The study of **Voronoi domains** helps quantify the effects introduced by these spaces on the theoretical predictions of CMB anisotropies. Which is important to consider when comparing theoretical predictions in models of finite cosmic topology with observational data.

Finally, we discussed statistical tests for **CMB isotropy**, focusing on the decomposition of temperature fluctuations into spherical harmonics. The power spectrum C_ℓ should be **statistically isotropic** in a homogeneous infinite universe, but anomalies in the **low multipole moments** ($221 \leq \ell \leq 240$) **suggest possible deviations from isotropy**. By analyzing the test statistic S_i across different frequency bands (Q, V, W from WMAP), we find that these anomalies persist even after accounting for instrumental noise, implying a possible **cosmological origin rather than observational artifacts**. Reinforcing the idea of the universe having a non-trivial cosmic topology, different from the one proposed by the Λ CDM model.

While no topology can be confirmed definitively, the mathematical and observational evidence discussed in this paper supports the idea that the universe may possess a non-trivial, multiconnected topology. Furthermore, finite spherical models remain viable candidates for explaining observed CMB anomalies. However, in order to reach a definitive answer, more precise CMB measurements and improved statistical analysis are required. Future CMB experiments,

such as higher-resolution observations from Planck and next-generation satellite missions, along with further advancements in computational methods and theory, will be beneficial in testing these hypotheses more rigorously and improve our understanding of the topology of the universe. Further research could focus on developing statistical tests for CMB isotropy in an inhomogeneous universe. An important area for future research is the development of statistical tests for inhomogeneous spaces. Since, while significant progress has been made in analyzing homogeneous spherical manifolds, inhomogeneous spaces introduce observer-dependent effects that complicate standard isotropy tests.

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